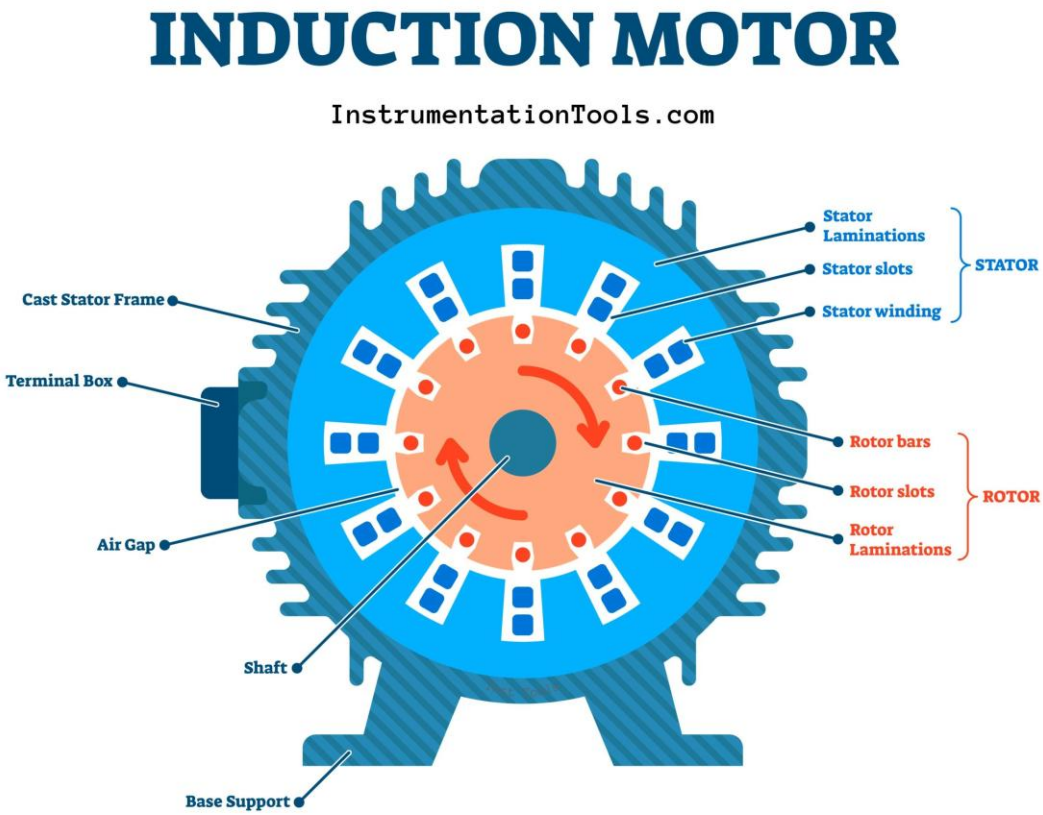
Exp 3

Induction Motor



Name: Abdelrahman Sherif

Code: 2300100 Sec: 5

Spring 25

Induction Motor Construction:

Stator: The stationary part of the motor that produces a rotating magnetic field when connected to an AC supply.

Rotor: The rotating part placed inside the stator. It is acted upon by the rotating magnetic field and starts to rotate.

Air Gap: A small clearance between the stator and rotor that allows smooth rotation.

Bearings: Support the rotor and reduce friction during rotation.

Cooling System: Often includes fans or cooling fins to dissipate heat.

Working Principle:

Rotating Magnetic Field Generation: When AC voltage is supplied to the stator, it creates a rotating magnetic field inside the motor.

Induced EMF and Current: The rotating magnetic field cuts across the rotor conductors, inducing an EMF according to Faraday's Law, and causing current to flow in the rotor.

Torque Production: Due to the interaction between the induced rotor current and the stator magnetic field, torque is produced according to Lorentz force law, causing the rotor to turn.

Slip: The rotor never reaches the synchronous speed but runs slightly slower. The difference between synchronous speed and rotor speed is called slip.

Efficiency Considerations: Losses include copper losses (I^2R in stator and rotor), core losses (hysteresis and eddy current losses), and mechanical losses (friction and windage).

No-Load Characteristics

Circuit Setup: Connect the motor to a rated voltage supply with no mechanical load on the shaft. Use proper measuring instruments.

Initial Operation: Energize the stator and allow the motor to reach steady-state speed.

Voltage, Current, and Power Measurement: Record the stator current and input power at no-load condition.

Core and Friction Losses: At no-load, the measured power mainly represents core losses, friction, and windage losses.

Reduced Voltage Testing: Repeat measurements with 80% of the rated voltage and observe changes in no-load current and power.

During Load Test, explain what will happen to the stator current when mechanical load increases on the motor shaft:

Increased Rotor Current: As the mechanical load increases, the rotor slows down slightly, increasing the slip.

Higher Stator Current: To maintain torque, the motor draws more current from the supply.

Voltage Regulation: There might be a slight voltage drop due to increased current.

Increased Copper Loss: Higher current leads to more I^2R losses in the stator and rotor windings.

Lab Readings

	No-Load Test	Locked Rotor Test
Voltmeter	59 V	132.5
Ammeter	0.33 A	3.5 A
Wattmeter	10 W	170 W

